Credit Risk: Part I

Lecture 5

BANK3011

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Objectives of this week's lecture

- To examine different types of loans extended and the nature of Credit Risk in banks

- To examine techniques used to measure specific borrower credit risk
  - ✓ Traditional Models
    - ➢ Qualitative approaches
    - ➢ Credit scoring
  - ✓ Newer Models (Market based models)
    - ➢ Interest rate spreads
    - ➢ RAROC

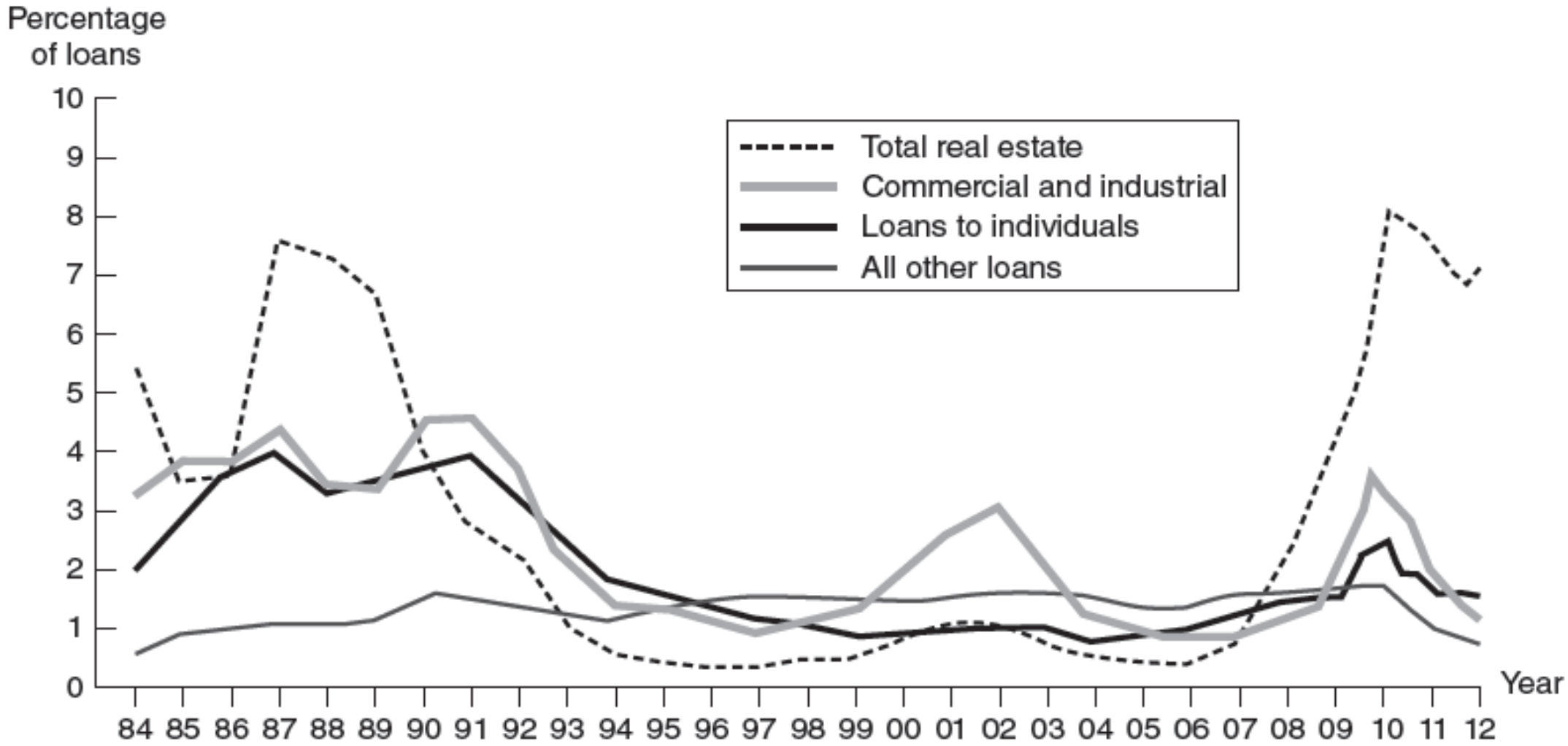
1. Credit risk

- The potential that an obligor will fail to meet its obligations in accordance with agreed terms.
- The risk of default, including cross-default
- The risk that security supporting the asset will be insufficient to cover any losses arising from default
- Opportunity gains or losses arising from changes in credit quality

- **Traditionally credit risk assessment has focused on default as the only credit event, modern credit risk approaches recognise that other credit events also create credit risk because they also affect the loan's value.**

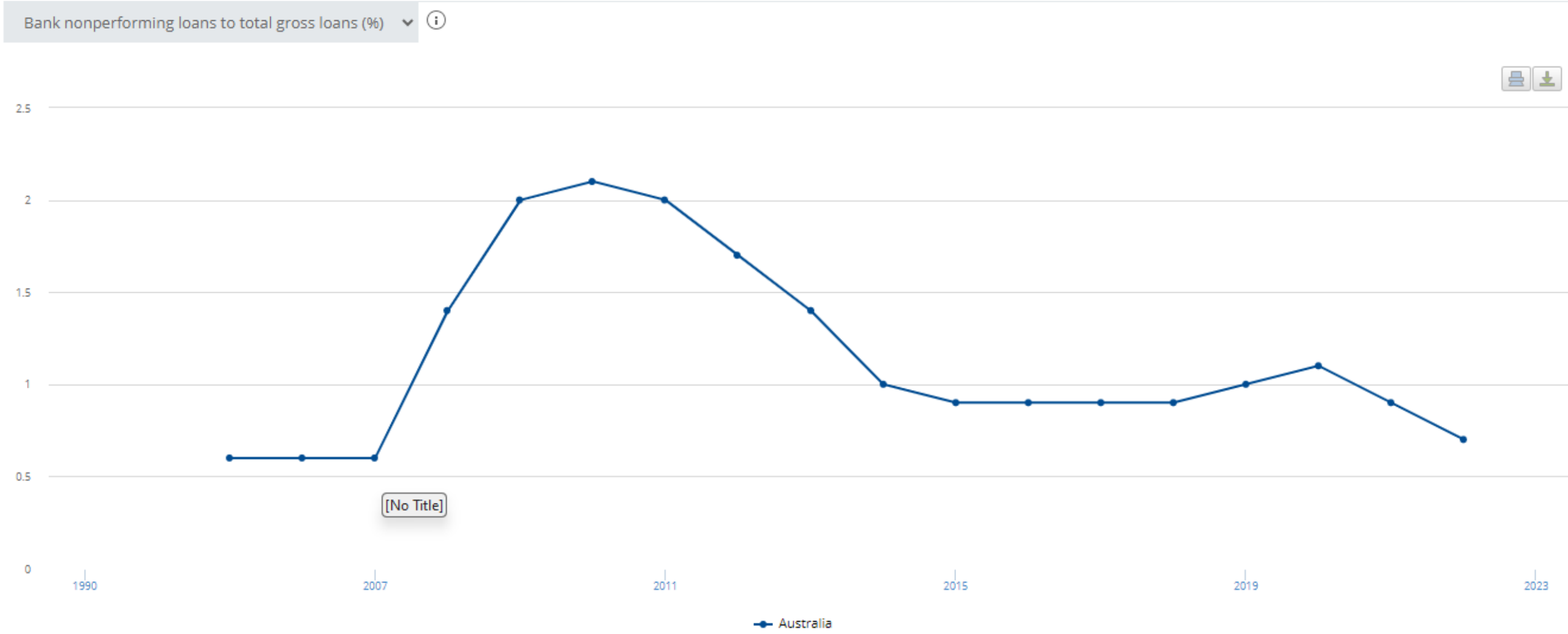
1. Credit risk

- Nonperforming Asset Ratio for U.S. Commercial Banks



1. Credit risk

- Nonperforming Asset Ratio for Australian Banks (from World Bank)



1. Credit risk

- Credit Quality Problems
  - ✓ Over the early to mid 1990s, improvements in NPLs for large banks and overall credit quality
  - ✓ Late 1990s and early 2000s: Telecom companies, tech companies
  - ✓ Mid 2000s, economic growth accompanied by reduction in NPLs
  - ✓ Mortgage crisis 2007-2008
  - ✓ Increased emphasis on credit risk evaluation

1. Credit risk

- Types of Loans
  - ✓ Commercial and industrial loans,
  - ✓ Real estate loans,
  - ✓ Individual (consumer) loans,
  - ✓ Other loans (e.g. government loans, margin loans, farm/agribusiness loans).

- Commercial and Industrial Loans
  - ✓ Loans for business purposes
    - ➢ Short-term loans: to finance working capital needs and other short-term funding needs,
    - ➢ Long-term loans: to finance purchase of real assets, new venture start-up costs and permanent increases in working capital.
  - ✓ Syndicated loans:
    - ➢ Finance provided by a group of lenders,
    - ➢ Usually to finance large C&I loans.

1. Credit risk

- Commercial and Industrial Loans
  - ✓ Credit risk:
    - ➢ Loan rating: external credit rating.
    - ➢ Secured (asset-backed) loan: loan backed by first claim on certain assets.
    - ➢ Unsecured loan (junior debt): loan with a general claim only.
  - ✓ Other characteristics
    - ➢ Fixed rate or floating rate
    - ➢ Spot loan or loan commitment (line of credit) which is a revolving credit facility with a maximum size and a maximum period of time

1. Credit risk

- Commercial and Industrial Loans
  - ✓ Commercial paper and disintermediation
    - ➢ Commercial papers: unsecured short-term debt instrument
    - ➢ Large corporations with good credit rating can get direct access to funds in capital markets by issuing commercial papers.
  - ✓ Changes:
    - ➢ Decline in C&I loans originated by commercial banks and growth in commercial paper market
    - ➢ Effect of financial crisis on commercial paper market

1. Credit risk

- Real Estate Loans
  - ✓ Loans to individuals and families to finance property purchase.
  - ✓ Long maturities, often 25 - 30 years.
  - ✓ Periodic payments typically cover both interest payments and partial repayment of principal.
  - ✓ Changing trend: in the 1990s most real estate loans were granted for owner-occupation; today a larger proportion of loans are for investment purposes.
  - ✓ Competitive market through new entrants
  - ✓ Fixed-rate, ARM
  - ✓ Mortgages can be subject to default risk when loan-to-value rises

1. Credit risk

- Individual (Consumer) Loans
  - ✓ Personal loans, automobile, mobile home loans
    - ➢ Non-revolving loans
  - ✓ Credit cards
    - ➢ A revolving loan facility
    - ➢ Growth in credit card debts
    - ➢ Risks affected by competitive conditions and usury ceiling

- Other loans
  - ✓ Other loans include a wide variety of borrowers and types
    - ➢ Farm (Agribusiness) loans
    - ➢ Other banks
    - ➢ Nonbank FIs: Broker margin loans
    - ➢ Foreign banks and sovereign governments
    - ➢ State and local governments

- Calculating the Gross Return on a Loan (Return on Asset approach)
  - ✓ Factors affecting the promised loan return:
    - ➢ Loan interest rate = base lending rate (BR) + credit risk premium (Φ).
    - ➢ Direct fees, eg. loan origination fee (of),
    - ➢ Indirect fees, eg.
      - ▪ Compensating balance requirements (b),
      - ▪ Reserve requirement (RR).
    - ➢ Other non-priced items, eg. collateral,
  - ✓ Gross return on loan (k) per dollar lent:

$$1+k = 1 + [of + (BR+ \Phi)] / [1- b(1-RR)]$$

1. Credit risk

- Calculating the Gross Return on a Loan – sample question
  - ✓ Suppose that a bank does the following:
    - ➢ 1. Sets a loan rate on a prospective loan at 8 percent (where BR = 5% and ϕ = 3%).
    - ➢ 2. Charges a 1/10 percent (or 0.10 percent) loan origination fee to the borrower.
    - ➢ 3. Imposes a 5 percent compensating balance requirement to be held as noninterest-bearing demand deposits.
    - ➢ 4. Pays reserve requirements of 10 percent imposed by the Federal Reserve on the bank's demand deposits.
  - ✓ Calculate the bank's ROA on this loan.

- Calculating the Gross Return on a Loan – an example

  ✓ Calculate the bank's ROA on this loan.

$$1 + k = 1 + \frac{0.0010 + (0.05 + 0.03)}{1 - [(0.05)(0.9)]} = 1 + \frac{0.081}{0.955} = 1.0848 \text{ or } k = 8.48\%$$

- Calculating the Expected Return on a Loan
  - ✓ Expected return might differ from promised return because of default risk.
  - ✓ Expected return:

$$E(1+r)=1+E(r) = p(1+k)+(1-p)*0 = p(1+k).$$

Where:

p = probability of loan repayment.

  - ✓ If $p < 1$:
    - ➢ Default risk exists

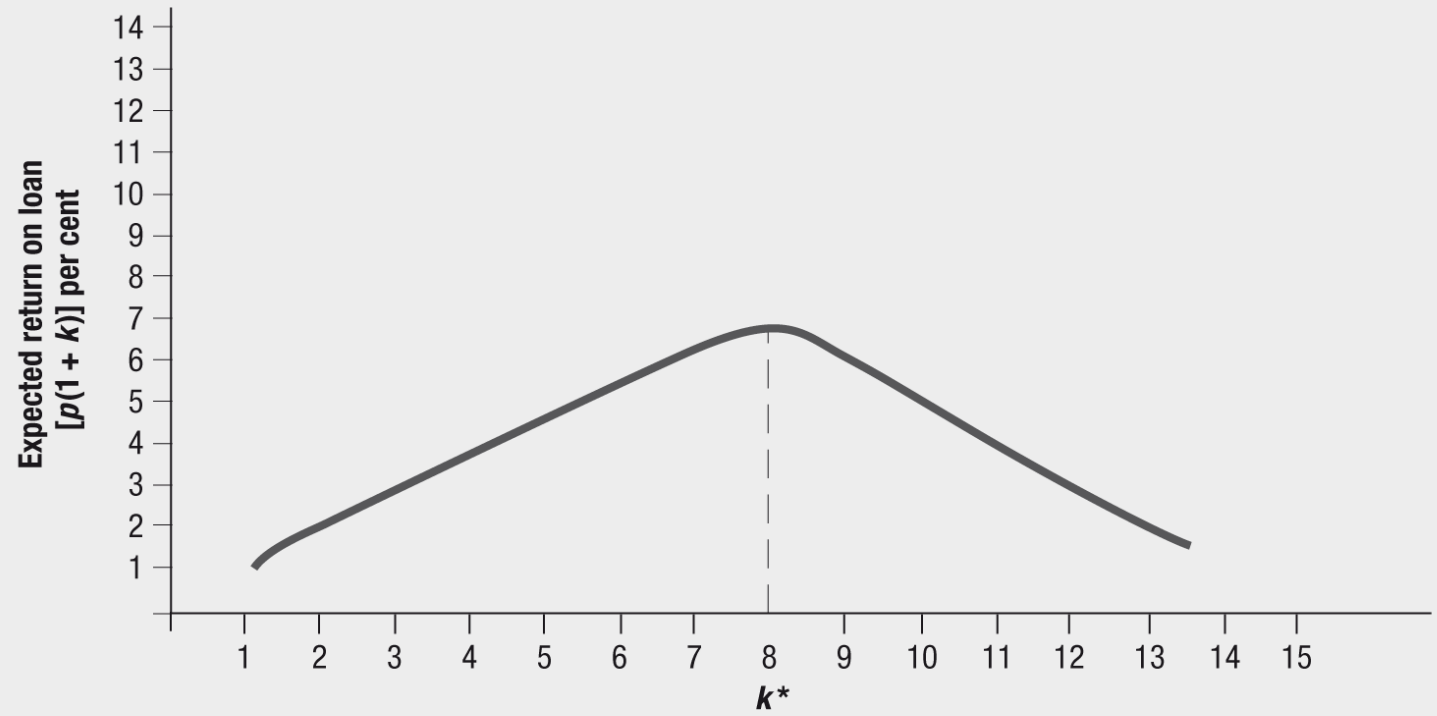
- Retail Versus Wholesale Credit Decisions
  - ✓ Retail:
    - ➢ Characteristics
      - ▪ Small dollar size loans,
    - ➢ Decision:
      - ▪ Usually accept or reject decision,
      - ▪ Standard loan rate is usually charged,
    - ➢ Credit risk controlled through credit rationing: limit the total exposure to the sector/individual

- Retail Versus Wholesale Credit Decisions
  - ✓ Wholesale:
    - ➢ Credit risk controls:
      - the use of differential interest rates to compensate for different levels of risks
      - Limit credit exposure through credit rationing.
    - ➢ Recall: expected return on loan $E(1+r)= p(1+k)$
      - Pricing tool (increase k) does not fully address the problem of credit risk
      - Why?

- Figure 10-4 in SC, 8°

$$1+E(r) = p(k)(1+k)$$

FIGURE 8.5 The relationship between the promised loan rate and the expected return on the loan



2. Estimating credit risk

- Credit Risk (like all risks) is viewed from the perspective of the risk of loss
- Credit risk can therefore be seen as the risk of the decline in the value of a credit asset as a result of unfavourable credit events including:
  - ✓ Default and;
  - ✓ Deterioration in creditworthiness (with default being the ultimate stage in this process of deterioration)

2. Estimating credit risk

- Traditional approaches:
  - ✓ Qualitative credit risk models
  - ✓ Credit scoring models

- Newer approaches:
  - ✓ Term structure based methods
  - ✓ Mortality rate models
  - ✓ RAROC models

2. Estimating credit risk

- Qualitative credit risk models
  - ✓ Borrower-specific factors include:
    - ➢ Reputation: implicit contract,
    - ➢ Leverage (capital structure),
    - ➢ Volatility of earnings,
    - ➢ Covenants,
    - ➢ Collateral.
  - ✓ Market-specific factors include:
    - ➢ Business cycle,
    - ➢ Level of interest rates.

2. Estimating credit risk

- Credit Scoring Models
  - ✓ Quantitative models that use observed borrower characteristics that are indicators of default to:
    - ➢ Calculate a score as a proxy of borrower's default probability, or
    - ➢ Sort borrowers into different default classes.
  - ✓ A scoring model needs to:
    - ➢ Establish factors that help to explain default risk
    - ➢ Evaluate the relative importance of these factors

2. Estimating credit risk

- Credit Scoring Models
  - ✓ These models are often used as
    - ➢ support for qualitative analysis for large loans or
    - ➢ to approve smaller retail loans
  - ✓ Major types of credit scoring models:
    - ➢ 1) Linear probability model; 2) Logit model & 3) Linear discriminant models

2. Estimating credit risk

- Linear Probability Model

$$Z_i = \sum_{j=1}^{n} \beta_j X_{i,j} + \text{error}$$

  - ✓ Major weakness that the estimated probabilities of default might lie outside the interval of zero and one.

- Logit Model
  - ✓ overcomes the weakness of the linear probability model using a transformation (logistic function) that restricts the probabilities to the zero–one interval.

$$P_i = \exp(Z_i)/(\exp(Z_i)+1)$$

- Credit Scoring Models: Linear Discriminant Models

  ✓ Altman's Z score model for manufacturing firms

$$Z=1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

Critical values of Z = 1.81 and Z = 2.99.

X_1 = Working capital/total assets.

X_2 = Retained earnings/total assets.

X_3 = EBIT/total assets.

X_4 = Market value equity/book value LT debt.

X_5 = Sales/total assets.

2. Estimating credit risk

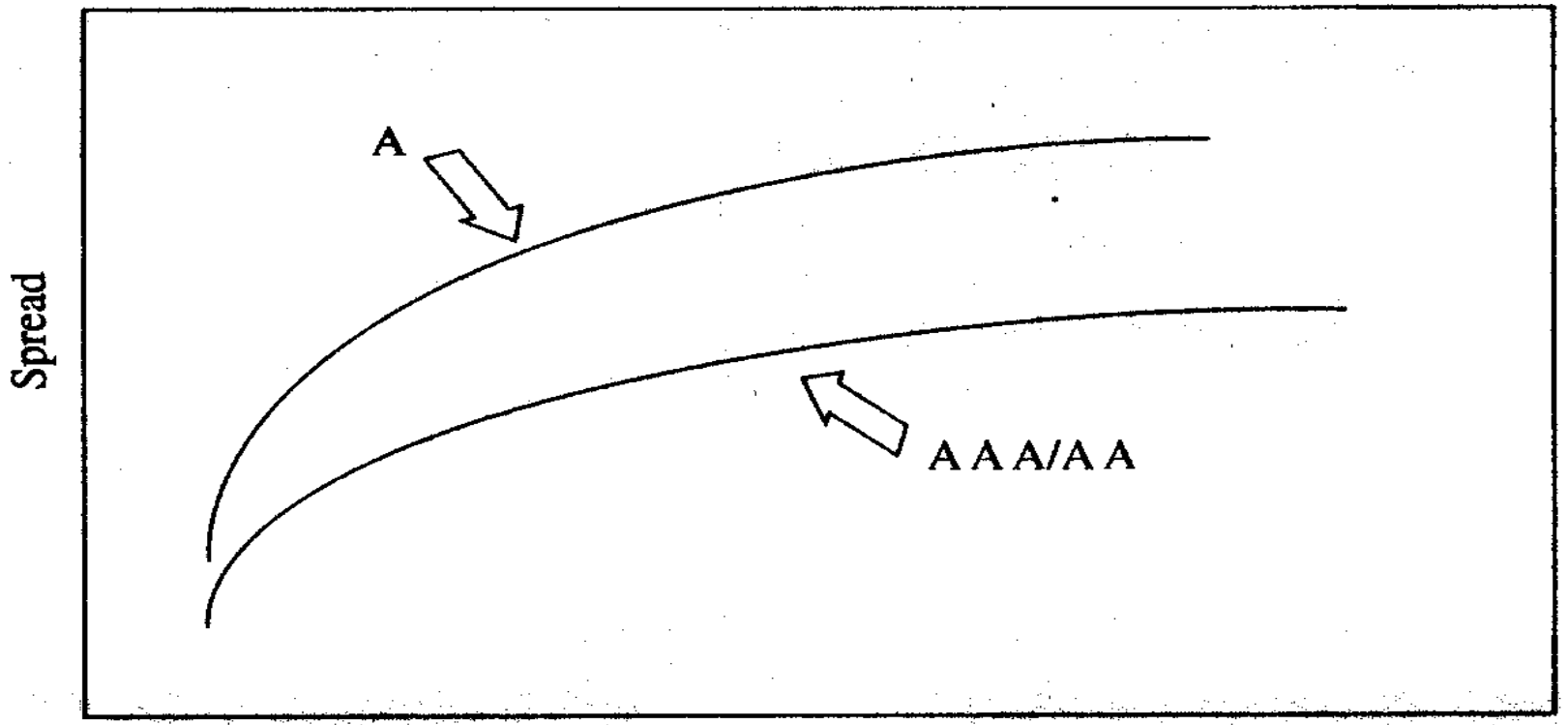
- Credit Scoring Models: Linear Discriminant Models
  - ✓ Critical values suggested by Altman
    - ➢ Z >= 2.99, high quality loans, low default risk
    - ➢ Z < 1.81, very low quality loans, high default risk
    - ➢ 1.81 <= Z < 2.99, hybrid

- Weaknesses:
  - ✓ Models ignore qualitative factors such as borrower reputation,
  - ✓ Variables in any credit scoring model unlikely to be constant over longer periods of time,
  - ✓ Weights in any credit scoring model unlikely to be constant over longer periods of time,
  - ✓ There is no centralised database on defaulted business loans for proprietary or other reasons – hard to test the validity of any model or develop new models.
  - ✓ Broad distinction between borrower categories, i.e. good and bad borrowers.

2. Estimating credit risk

- Term Structure of Credit Risk
  - ✓ Credit Spreads
    - ➢ The margin compared with the risk free rate designed to compensate the lender or investor for the risk of default
    - ➢ A market driven interest rate spread that indicates issuer specific credit risk

Figure 6.6 Term Structure of Credit Spreads



Source: Satyajit Das *Credit Derivatives: CDOs & Structured Credit Products*

- Term Structure of Credit Risk

$$C = k - i$$

Where:

C = Credit Spread

k = Yield on Credit Risky Security

i = Yield on Corresponding Risk Free Security

- Credit Spreads and Credit Risk
  - ✓ The Rational Investor will set returns as:

$$(1+i) = p(1+k) + (1-p) * 0$$

$$= p(1+k)$$

Where: p = Probability of Repayment

i = risk free rate

k = rate on credit risky asset

Note (1-p) = is therefore the probability of default

- Credit Spreads and Credit Risk
  - ✓ Default & Required Return

$$p = \frac{(1+i)}{(1+k)}$$

Where p is the probability of repayment

In addition however in the event that the obligor defaults the bank will receive some payment (*A*) by way of a recovery

- Credit Spreads and Credit Risk
  - ✓ Default & Required Return – sample question
  - ✓ A BBB Corporate debt has yield of 7.5%. There is a Treasury with very similar term but with yield of 3%. Please calculate the implied default probability of the BBB corporate debt.

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  - ✓ Default & Required Return – sample question
  - ✓ A BBB Corporate debt has yield of 7.5%. There is a Treasury with very similar term but with yield of 3%. Please calculate the implied default probability of the BBB corporate debt.

$$p = \frac{(1 + 0.03)}{(1 + 0.075)} = 0.9581$$

Hence, the default probability is $1 - p = 0.0419$ or 4.19%

- The Required Return incorporating recovery in the event of default
  - ✓ The formula can be expanded as:

$$1+i=[(1-p)\gamma(1+k)]+[p(1+k)]$$

  - ➢ (1-p) γ(1+ k) = the payoff the FI expects to get in the case of default.

- So What?
  - ✓ The above equation shows the relationship between the variables thus we can calculate one of the variables if the other two are known (or can be estimated):
    - ➢ Given default probabilities and the recovery rate we can calculate the implied spread
    - ➢ Given default probabilities and spread we can calculate the implied recovery rate
  - ✓ Importantly however for this topic:
    - ➢ Given the credit spread and recovery rate we can calculate the implied default probability

- Mortality-Rate Models
  - ✓ Essentially a marginal probability of default concept. The probability of default is estimated from past data on defaults.
  - ✓ Marginal Mortality Rates:

$$MMR_t = \textit{(Total value of bond/loan defaults in yr t of issue) / (total value of bond/loan outstanding in yr t of issue)}$$

  - ✓ Cumulative Mortality Rate is just the cumulative probability of default.

$$CMR_2 = 1 - (1-MMR_1)*(1-MMR_2)$$

  - ✓ Has many of the problems associated with credit scoring models, such as sensitivity to the period chosen to calculate the MMRs

- RAROC Models
  - ✓ RAROC = Risk-Adjusted Return On Capital
  - ✓ Essential idea: modified ROA concept which links expected interest income on a loan to the expected loan risk.

$$\text{RAROC} = \frac{\text{one year income on a loan}}{\text{loan}\left(\text{asset}\right)\text{risk or capital at risk}}$$

- RAROC Models
  - ✓ Loan approval if RAROC > benchmark return on capital, eg. ROE
  - ✓ Net income = value of loans outstanding * all-in-spread (AIS)
  - ✓ Where AIS = contractually-promised return on the loan (taking into account of interest rate spread and all explicit and implicit fees) – funding cost (spread + fees)

2. Estimating credit risk

- RAROC Models: Measuring loan risk
- Loan approval if RAROC > benchmark return on capital, eg. ROE
  - ✓ Capital at risk (VaR approach): the potential loan loss under adverse credit scenarios (see example 10-7)

$$\Delta L = - D * L * [\Delta R / (1+R)]$$

  - ➢ *D is the duration of the loan*
  - ➢ *L is the size of the loan*
  - ➢ *R is the current level of rate*
  - ➢ *ΔR is the increase in risk premium*

3. Others

- RAROC Models: Measuring loan risk
- Loan approval if RAROC > benchmark return on capital, eg. ROE
  - ✓ Capital at risk (VaR approach): the potential loan loss under adverse credit scenarios (see example 10-7)

under adverse credit scenarios,

such as 95 or 99 percentile value of the distribution

RAROC = One year income on loan / ΔL

3. Others

- RAROC Models: Measuring loan risk – sample question
- The duration of a soon to be approved loan of $10 million is four years. The 99th percentile increase in risk premium for bonds belonging to the same risk category of the loan has been estimated to be 5.5 percent.

1. What is the capital (loan) risk of the loan if the current average level of interest rates for this category of bonds is 12 percent?
2. If the fee income on this loan is 0.4 percent and the spread over the cost of funds (2 percent) to the bank is 1 percent, what is the expected income on this loan for the current year?

3. Others

- RAROC Models: Measuring loan risk – sample question
- The duration of a soon to be approved loan of $10 million is four years. The 99th percentile increase in risk premium for bonds belonging to the same risk category of the loan has been estimated to be 5.5 percent.

3. What is the estimated risk-adjusted return on capital (RAROC) of this loan?

4. If the minimum RAROC acceptable to the bank is 8 percent, what should be its expected percentage fee income in order for it to approve the loan?

- RAROC Models: Measuring loan risk – sample question

1. What is the capital (loan) risk of the loan if the current average level of interest rates for this category of bonds is 12 percent?

$$\Delta N = -D \times LN \times \frac{\Delta R}{1+R}$$

$$\Delta N = -4.0 \times \$10{,}000{,}000 \times \frac{0.055}{1.12} = -40{,}000{,}000 \times 0.049107 = -\$1{,}964{,}280$$

- RAROC Models: Measuring loan risk – sample question

2. If the fee income on this loan is 0.4 percent and the spread over the cost of funds to the bank is 1 percent, what is the expected income on this loan for the current year?

Fees = 0.004 × 10,000,000 = $40,000

Spread = 0.01 × 10,000,000 = $100,000

One-year net income on loan = $140,000

- RAROC Models: Measuring loan risk – sample question

3. What is the estimated risk-adjusted return on capital (RAROC) of this loan?

RAROC = (one-year net income on loan) ÷ (loan risk)

RAROC = $140,000 ÷ $1,964,280 = 0.071

4. If the minimum RAROC acceptable to the bank is 8 percent, what should be its expected percentage fee income in order for it to approve the loan?

RAROC = [(spread × loan amount) + (fee × loan amount)] ÷ (loan risk)

0.08 = [100,000 + (fee × 10,000,000)]÷ (1,964,280)

((0.08 × 1,964,280) - 100,000)) ÷ 10,000,000 = fee = 0.005714 or 0.5714%.

Summary

- There are different types of loans extended by FIs

- It is important to analyse and measure credit risk on individual loans

- Important for the purposes of:
  - ✓ Acceptance/Rejection decisions
  - ✓ Pricing loans and bonds
  - ✓ Setting limits on credit risk exposure